

Algebraic manipulation



1

a $x = 5, -5$

b $x = \pm 7\sqrt{2}$

c $x = \frac{8}{9}, \frac{-8}{9}$

d $p = \pm 12$

e $x = 0, \frac{12}{7}$

f $x = 4 \pm \sqrt{10}$

g $x = \frac{5}{4}, -\frac{11}{4}$

h $x = \frac{9}{4}$

i $x = 1$

j $m = 0, 10$

2

$$x = \pm \sqrt{\frac{c}{a}}$$

Factorisation

3

a $x = 0, -12$

b $y = 0, \frac{1}{5}$

c $m = 0, 14$

d $x = -10, -9$

e $m = 5, -4$

f $x = 10$

g $x = -4, 5$

h $m = 14, 13$

i $x = 8, -3$

j $y = -1, 3$

k $x = 2, -4$

l $x = \frac{9}{4}, 4$

m $x = -\frac{5}{4}, -3$

n $x = -\frac{4}{5}, -2$

o $x = \frac{5}{4}, 3$

p $x = -\frac{5}{3}, 4$

q $x = \frac{-5}{2}, 12$

r $b = -\frac{5}{3}, \frac{3}{4}$

s $k = 1, \frac{7}{6}$

4 $x = \pm 4$

5 $x = 2$

6

a $(x - 7)(x + 4) = 0$

b $x(x + 1) = 0$

7 $x^2 + 8x + 15 = 0$

Using technology

8

a $x = 2, -\frac{5}{3}$

b $x = \frac{3}{2}, -\frac{7}{2}$

c $x = 0.4, -2.0$

9

a $x = \pm\sqrt{2}$



b $x = 1.4, -1.4$

10

a 1

b 1

Applications

11 None

12 $x = \frac{12 \pm \sqrt{1044}}{90}$

13 $x = -\frac{3}{5}, -2, -\frac{5}{2}, -4$

14 $k = -16$

15 $a = 13$

16 $t = 22, 14$

17

a $t = 4, 8$

b 4 hours

c 4 hours

18

a $x = 0, 32$

b 32 cm

19

a $17 + \frac{6426}{x^2} = \frac{663}{x}$

b $x = 18, 21$

c 36.83 cm

20

a 0, 14 months

b No, after one year the company will still be spending more money than it is making.